Network Centrality and Lead-Lag Return Predictability in US Tech/Semiconductor Stocks

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1. Hypothesis Pitch

Some stocks are watched more closely than others, and when one of these highly connected ‘hub’ stocks makes a sharp move, their less-watched but economically linked peers do not react straight away but instead catch up over the following days.

This informs the strategy: watch hub stocks for a sharp move, then buy (or short) their connected peers in the same direction, aiming to make a profit as those peers catch up to the ‘hub’.

The mechanism is attention-driven, not purely statistical. Analysts, passive index flows, and algorithmic trading concentrate disproportionately on the most liquid, most-covered names. When genuine new information arrives — an earnings surprise, a demand shock, a capacity announcement — it is priced into these heavily-watched names almost immediately. Their economically linked peers share real exposure to the same information, through overlapping end demand, shared input costs, or a common sector narrative, but draw far less continuous monitoring, so their price adjustment should lag. This is a specific instance of the gradual information diffusion / limited-attention literature (Hou and Moskowitz, 2005, on price delay; Cohen and Frazzini, 2008, on economic links and predictable returns), not a generic trend-following story.

I chose the US tech/semiconductor sector specifically because it combines a small number of extremely heavily-covered mega-cap names with a long tail of smaller, less-watched but economically-linked peers — equipment makers, smaller analog and EDA firms — exactly the structural contrast the hypothesis needs, and a sector I follow closely.

What would prove this wrong: connected peers moving on the same day as the hub with no measurable lag, rather than after it; a genuine lag that fails to produce a positive risk-adjusted return once realistic transaction costs (5–10bps round-trip) are applied, especially out-of-sample; or the apparent “lag” turning out to be explained entirely by ordinary same-day correlation rather than a real catch-up dynamic — tested directly by comparing same-day versus later-day neighbour returns.

2. Data Pipeline

This strategy needs a universe of US tech and semiconductor stocks with enough shared economic exposure to actually form a network. Therefore 65 tickers were selected across mega cap technology names, semiconductor manufacturers and equipment makers, and smaller cloud and software companies, alongside the S&P 500 as a broad market benchmark.

Daily adjusted close prices were pulled from Yahoo Finance starting January 2019, giving us almost 8 years of history, which is enough for a 60-day rolling correlation window, an in-sample design period, and a genuine out of sample holdout.

The download is cached locally so that every run of the notebook uses an identical dataset, which is very important for reproducibility.

One of the tickers, JNPR, returned no data at all and was therefore dropped automatically, leaving 64 usable series plus the benchmark.

The raw data still needed cleaning before it could be used in the strategy. Adjusted close prices are used throughout rather than raw close, so stock splits and dividend payments do not appear as artificial price jumps or crashes in the return data.

Any stock missing more than five percent of its trading days over the sample was dropped entirely rather than patched over. Most of that missing data is not random: stocks like ABNB, ALGM, PLTR, SNOW, DDOG, NET and CRWD simply did not exist as public companies when the sample starts in 2019, since they only listed on the stock market afterwards. Filling in years of history for a stock that was not even trading yet would mean inventing price data rather than cleaning it, so these stocks are removed instead. QRVO is a different case: despite being a long-established company, it has almost no usable price data on Yahoo Finance at all within the sample period, which looks like a data provider issue rather than a real gap in its trading history, so it is removed for the same practical reason even though the cause is different. In total, 8 tickers were removed this way, leaving a clean universe of 56 stocks.

The stocks that stayed in can still have a handful of small, isolated gaps, usually just a day or two here or there from a data vendor hiccup rather than a missing listing. These gaps are filled in, but only under two conditions: the gap has to sit inside the stock's existing data, not before its first trading day or after its last, and it can be no more than five days long. Anything wider, or anything outside a stock's real trading history, is left as missing rather than guessed at, so no stock's price history is stretched beyond what Yahoo Finance actually recorded for it.

This universe reflects today's current set of tech and semiconductor names, not a point in time reconstruction, so any company that was delisted, acquired, or went bankrupt earlier in the sample period is absent by construction. That is a real limitation, and it is addressed directly in the Pitfall Checklist rather than glossed over here.

3. Strategy and Backtest

3.1 Strategy and Backtest Mechanics

Every week, the strategy builds a correlation network over the stocks in the universe. Each pair of stocks is linked if the correlation between their daily returns, measured over the trailing 60 trading days, is above 0.5. This is rebuilt fresh every week so the network can track how relationships between stocks shift over time, rather than assuming today's links held true for the whole sample.

Comparing every pair of stocks like this costs on the order of n squared operations, since every stock must be checked against every other stock. For the 56 stocks in this universe that works out to 1,540 pairs a week, and across all 396 weekly networks in the sample that comes to 609,840 pairwise comparisons in total. The notebook times this directly at 3.6 seconds, so at this scale it is not a practical bottleneck.

We calculate degree centrality, the share of all other stocks a given stock is connected to in that week's network, in order to identify that week's hubs: the six most connected stocks. The stocks that actually turned out to be hubs most often were MPWR, KLAC, ENTG, AMAT, LRCX, ADI, CDNS, MCHP, AVGO and SNPS. These are semiconductor equipment, analog and EDA names, not the mega cap stocks like AAPL, MSFT or NVDA that the hypothesis expected to dominate. That result matters enough to get its own explanation in the Methodology Defence.

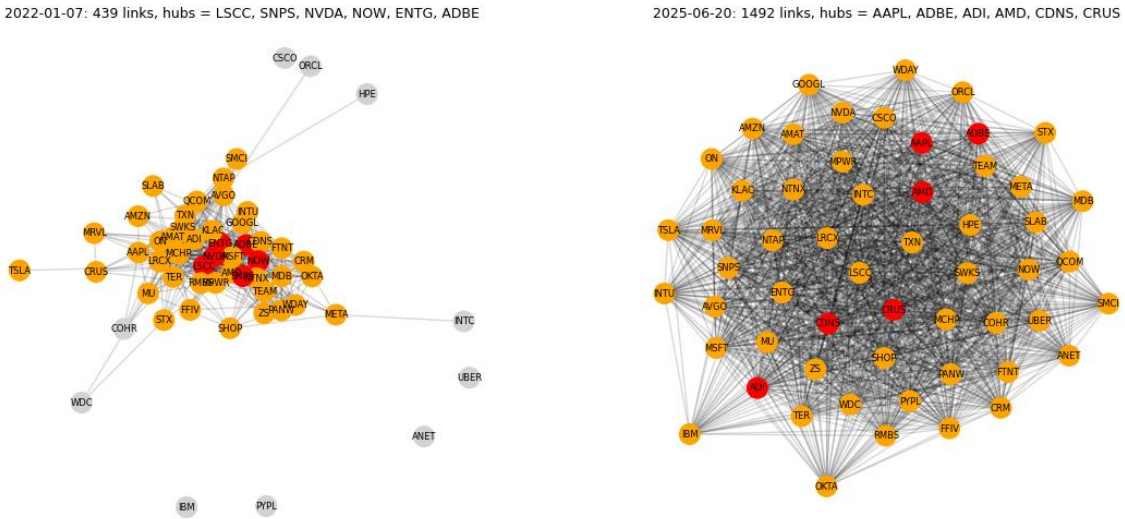


Figure 1 - Typical week versus a market-wide stress week (network snapshots)

In a typical week, such as 7 January 2022, 439 pairs are linked and almost the whole universe sits inside one dense cluster. That week's six hubs (LSCC, SNPS, NVDA, NOW, ENTG and ADBE) sit at the centre of it, while AAPL and AMZN are neighbours rather than hubs, and TSLA and META hang off the edge of the main group. Some names (CSCO, ORCL, HPE, IBM, PYPL, UBER, ANET, INTC, WDC and COHR) aren't linked to any hub that week.

This is the same pattern we reported above, where the hubs are semiconductor equipment, analog and EDA names, not the mega caps that our hypothesis expected to dominate.

A hub shock is flagged whenever a hub's return on a given day is more than twice as large as its own recent volatility. That volatility is measured only up to the previous day, so the signal never uses information from the day it is trying to predict. This produced 736 hub shocks across the whole sample: 639 of them in sample and 97 out of sample.

For each shock, the strategy buys an equal weighted basket of the hub's connected neighbours, or shorts it if the hub fell, at the next day's close. The position is held for three days, to give any catch-up time to actually happen. A trading cost of 10 basis points round trip is charged on every trade, and capital is split equally across whatever trades happen to be open at a given time, so the strategy never uses leverage. On average, a position was open on 48% of trading days.

3.2 Results

The six metrics compared across both tables are:

- **Cumulative return**: the total percentage gained or lost over the period.
- **Annualized return**: the cumulative return expressed as a compound annual rate, so that periods of very different lengths can be compared on equal footing. This goes beyond the brief's minimum.
- **Sharpe ratio**: return earned per unit of volatility taken on.
- **Maximum drawdown**: the worst peak to trough loss along the way.
- **Sortino ratio**: like the Sharpe ratio, but it only penalises downside volatility rather than volatility in either direction. This also goes beyond the brief's minimum.
- **Win rate**: the share of days with an open position that were actually profitable. This also goes beyond the brief's minimum.

Two benchmarks are included alongside the strategy in both tables: an equal weighted holding of all 56 stocks in the universe, held for the whole period, and the S&P 500 as a broad market benchmark. Both show whether the strategy actually adds anything beyond simply holding stocks.

3.2.1 In Sample Comparison

In sample, the full comparison looks like this:

Series	Cumulative Return	Annualised Return	Sharpe Ratio	Max Drawdown	Sortino Ratio	Win Rate
Strategy (with Costs)	-32.8%	-5.9%	-0.163	-57.1%	-0.148	49.8%
Strategy (No costs)	-12.7%	-2.1%	0.018	-49.9%	0.017	51.9%
Equal-weight stocks	424.3%	28.8%	0.969	-40.8%	1.327	55.0%
S&P500	135.1%	14.0%	0.745	-33.9%	0.908	54.8%

Table 1 - In-sample performance comparison.

This is also shown in the chart below:

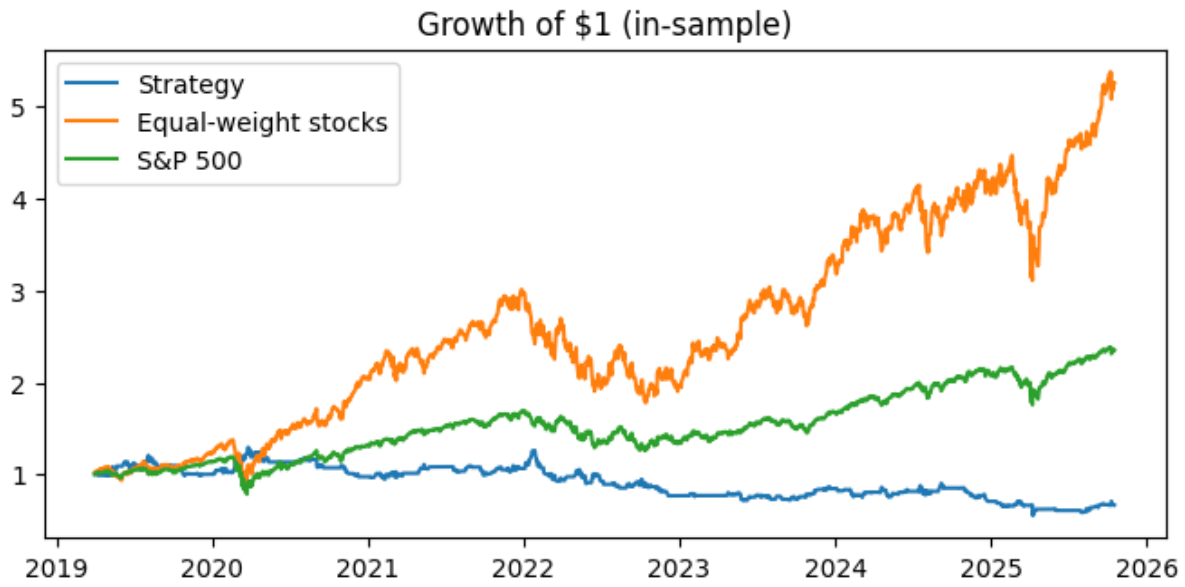


Figure 2 - Growth of $1, in-sample

The strategy's line sits close to flat for the full six and a half years rather than trending steadily downward, consistent with being out of the market roughly half the time whereas both benchmarks compound upward across the whole window.

3.2.2 Out of Sample Comparison

Out of sample, the same comparison looks like this:

Series	Cumulative Return	Annualised Return	Sharpe Ratio	Max Drawdown	Sortino Ratio	Win Rate
Strategy (with Costs)	9.5%	9.5%	0.453	-18.2%	0.458	50.0%
Strategy (No costs)	14.1%	14.1%	0.590	-17.7%	0.595	50.4%
Equal-weight stocks	48.7%	48.9%	1.582	-14.6%	2.480	57.0%
S&P500	15.7%	15.7%	1.211	-9.1%	1.857	53.0%

Table 2 - Out-of-sample performance comparison.

This is also shown in the chart below:

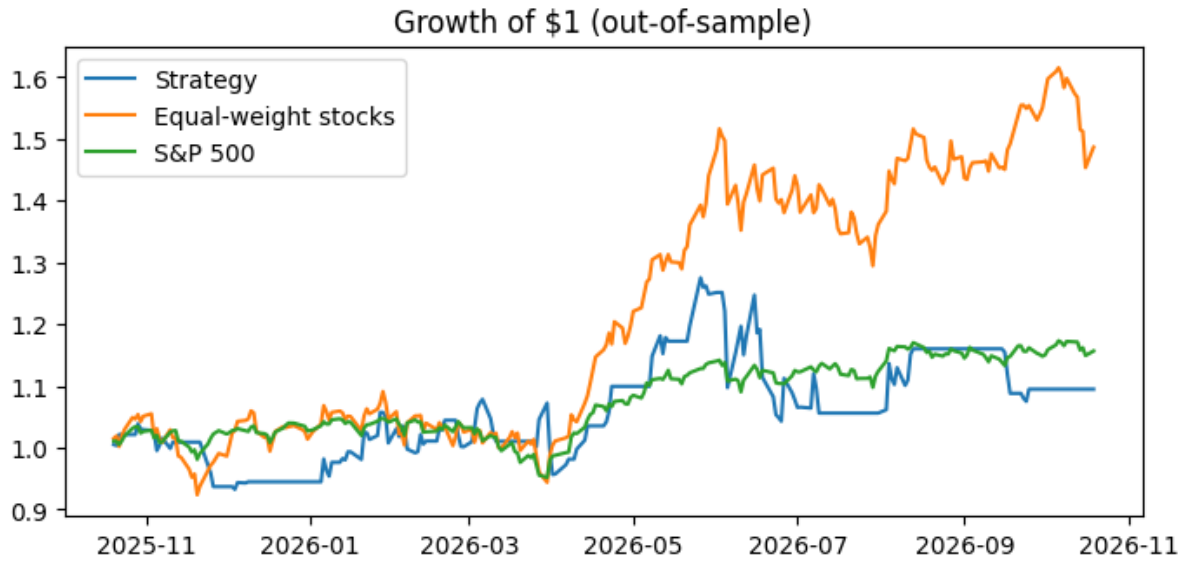


Figure 3 - Growth of $1, out-of-sample.

The strategy's line moves in visible steps rather than a smooth curve, a direct reflection of capital sitting in cash on days without an open trade, it tracks close to the S&P 500 for most of the window before the equal-weight basket pulls further ahead.

The in sample and out of sample cumulative return figures are not directly comparable on their own: the two periods cover very different lengths, around 6.5 years against the 1 year held out, so a larger magnitude in one period partly just reflects more time to compound. Annualized return puts both periods on the same yearly footing, and the Sharpe and Sortino ratios are already annualized for the same reason, which is why those three metrics, not raw cumulative return, are the right way to compare performance across the two periods.

Looking at both tables together, the pattern holds across every metric. The strategy's win rate sits right at 49.8% in sample and 50.0% out of sample, essentially a coin flip on the specific long or short bets it actually makes, which is the real test of whether those bets carry any skill. The Sortino ratio tells much the same story as the Sharpe ratio here, since the strategy's gains and losses are similar in size, but it still matters to report alongside it, since it isolates the downside specifically rather than treating all volatility as risk. Annualized return tells the same story as cumulative return once both periods are placed on the same yearly footing: the strategy returns -5.9% a year in sample and 9.5% a year out of sample, against 28.8% and 48.9% for the equal weight benchmark over the same two periods. Taken together, the three added metrics do not rescue the strategy: they confirm the same underperformance the required metrics already show, just on a basis that is actually comparable across the two periods.

Removing trading costs entirely improves both periods, but the in sample period improves the most: the Sharpe ratio moves from -0.163 to roughly flat at 0.018. That gap shows a meaningful part of the in sample loss comes from trading too often relative to the size of the edge, not purely from a weak signal.

As the tables show, both benchmarks comfortably beat the strategy on every metric, in both periods: higher Sharpe and Sortino ratios and smaller drawdowns. The honest conclusion is that this is a negative result. The lead lag strategy, built exactly as specified and locked before the backtest ran, does not beat simply holding the underlying stocks, in either period.

4. Methodology Defence

4.1 Pre-registered Parameters

Every parameter in this strategy, the 60 day window, the 0.5 correlation threshold, degree centrality as the measure of a hub, treating the six most connected stocks each week as hubs, the shock trigger of twice trailing volatility, and the 3-day hold, was fixed before the backtest ever ran. None of it was tuned afterwards to make the result look better.

That matters because the whole point of locking these choices in advance is to make the backtest's result trustworthy evidence, whichever way it comes out. A strategy that is tuned after seeing the data can be made to look good almost regardless of whether the underlying idea is real, so the only way to find out if the idea is real is to commit to a design first and then test it honestly.

4.2 Testing the Mechanism

The lag check in the Sanity Checks section is included here specifically to defend the backtest's own machinery, separate from whether the strategy actually makes money. It tests the mechanism the hypothesis depends on directly: do a hub's neighbours actually move with it, and with a lag?

Connected neighbours do move with the hub, and strongly: on average 380 basis points in the hub's direction on the shock day itself. But the day after, the average move is actually slightly negative, at -46 basis points, and days two and three only add back 27 and 8 basis points between them, shown in the chart below.

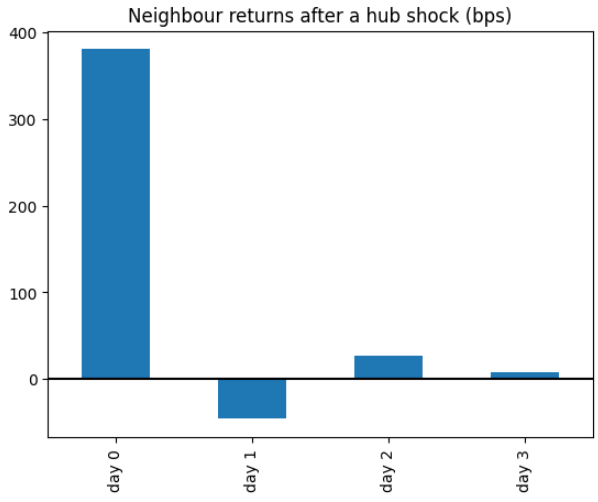


Figure 4 - Neighbour returns after a hub shock.

An actual catch-up would show the later bars growing to day 0’s size as the lag resolved however the move is instead absorbed almost entirely on the shock day itself.

There is essentially no multi day catch up to trade. What the network is picking up is same day movement together, not the gradual, lagged information diffusion the hypothesis pitch describes.

That distinction is what makes this a methodology defence rather than just a result: it shows the weak performance is a property of the hypothesis, not a flaw in the pipeline. If the lag check had shown a real catch up and the strategy had still lost money, that would point to a bug. It did not, so the result can be trusted as an honest test of the idea.

4.3 Threshold Sensitivity

0.5 was chosen as the correlation threshold because it sits in the range conventionally treated as a meaningful correlation: strict enough to link only pairs with a real relationship, loose enough that the weekly network does not collapse to almost no edges.

Rerunning the backtest at 0.4 and 0.6 defends that single choice directly. Out of sample, 0.4 produces a negative Sharpe ratio of -0.318, while 0.5 and 0.6 both come out positive, at 0.453 and 0.249.

0.5 sits on the stronger side of that range rather than in the middle of a clearly robust one, so the result should be read with some caution. It was still kept at 0.5, set before any of this was run, rather than moved to whichever threshold happened to look best afterwards.

At 0.5, the network neither collapses nor saturates in a typical week, sitting at a 28% median density, though it does nearly saturate in stress periods such as the week of 20 June 2025 (shown later in Figure 5).

4.4 Degree Centrality

Degree centrality was used to identify hubs because it is simple and transparent: a stock counts as a hub if it is directly correlated with a large share of the rest of the universe, with no weighting or recursive definition to justify. A more elaborate measure, such as eigenvector or betweenness centrality, or simply hand picking well known mega caps as the hubs, could have been used instead.

Degree centrality was kept because the whole point of the measure is to test whether connectedness and attention are the same thing, rather than assume the answer by picking the names expected to matter in advance.

The hubs that actually emerged, MPWR, KLAC, ENTG, AMAT, LRCX, ADI, CDNS, MCHP, AVGO and SNPS, are semiconductor equipment, analog and EDA names, not the mega cap stocks the hypothesis expected. That suggests degree centrality here may be picking up shared exposure to the semiconductor cycle and common input costs across a tightly clustered group of companies, rather than which stocks are most watched by analysts

and passive investors. The attention-based story may still be partly true, but a shared sector factor explanation looks at least as plausible.

This is not just a visual impression from the list of names. The same degree centrality measure, averaged across every weekly network in the sample, a calculation that reproduces the notebook's own hub counts exactly as a check, puts hard numbers behind it.

AAPL ranks 29th out of the 56 stocks, MSFT 18th, GOOGL 35th, AMZN 28th, META 45th, and TSLA 53rd, essentially the least connected stock in the entire universe. NVDA does better than the other mega caps, at 14th, but still outside the ten most connected names.

If degree centrality were tracking which stocks are most watched, the most famous, most heavily covered names should sit near the top. Instead, they sit in the bottom half, and in TSLA's case, near the bottom of the whole universe.

5. Pitfall Checklist

5.1 Look-ahead Bias

Look-ahead bias is directly addressed and accounted for within the notebook, not just something that was assumed. Take the very first shock in the sample and follow the exact pathway in order:

1. 29 March 2019: the weekly network is built using only the 60 days of returns available up to that date. This is what fixes MPWR as a hub for the week, and fixes which stocks are its connected neighbours.
2. 2 April 2019: MPWR's trailing volatility, the "normal" move size it is compared against, is measured using only data up to this date.
3. 3 April 2019: MPWR's actual return that day is checked against that volatility figure, and it is large enough to count as a shock.
4. 4 April 2019: the trade in MPWR's neighbour stocks is entered, one full day after the shock happened.

Every step only ever uses information that existed before it, in the exact order it would have been known in real time, therefore removing any look ahead bias from the signal. This is confirmed with an explicit check in the code that passes for the trade examined.

5.2 Survivorship Bias

The universe used here is today's current list of 56 tech and semiconductor stocks, not a reconstruction of what the sector actually looked like at each point between 2019 and 2026. That means anything that was delisted, acquired, or went under during that period is simply missing from the data.

JNPR is the clearest example. It disappears entirely because it was delisted after HPE acquired it in 2025, and it was dropped automatically before the cleaning step even ran. This matters because a stock that gets removed from a sector, almost always because it failed or merged away, tends to be different from the stocks that survived the whole period, and that difference is not captured anywhere in this analysis.

This most likely biases the results upward. The universe only contains companies that stayed public and relevant through to today, excluding failures by construction, so both the strategy and its equal weight benchmark trade within a universe of only survivors. Their absolute returns are therefore probably inflated, in a similar direction for both, so the finding that the strategy underperforms the benchmark holds up better than the raw return figures do on their own.

This is left as a known limitation and is not accounted for currently. A possible extension to the project that could rectify this would be by using a point in time constituents list, a

historical record of which tech and semiconductor stocks were actually tradeable at each date, including the ones that later failed or were acquired, rather than filtering the universe down from today's surviving names.

5.3 Overfitting/ Data Snooping

Every parameter this strategy depends on, the window, the threshold, the number of hubs selected each week, the hold period and the shock size, along with the hypothesis itself, was written down and fixed before the backtest ever ran. Nothing was adjusted afterwards based on how well it performed. This is the core protection against data snooping in this project: the design was committed to before the result was known, not shaped by it.

The threshold sensitivity check is a second line of defence on top of that. Rerunning the strategy at correlation thresholds of 0.4 and 0.6, alongside the locked 0.5, shows the result is not fully stable: the out of sample Sharpe ratio is negative at 0.4 but positive at 0.5 and 0.6. This is reported openly, in the Methodology Defence, rather than hidden, and it is a reason to treat the single positive result at 0.5 with some caution rather than full confidence.

5.4 Transaction Costs/Slippage

A cost of 10 basis points round trip is charged on every trade, a reasonable estimate for liquid tech and semiconductor stocks. This is still a simplification: the cost is applied once per trade rather than separately on entry and exit, and it does not account for slippage from trading the less liquid neighbour stocks in anything beyond small size.

Costs clearly matter here: removing them entirely moves the in sample Sharpe ratio from -0.163 to roughly flat at 0.018, a sign that the strategy is trading too often relative to the size of its actual edge.

5.5 Out-of-Sample Validation

The final 12 months of the sample were held out from the entire design process and used only once, at the end, to test the strategy exactly as it was locked. Out of sample, the strategy returned 9.5%, with a Sharpe ratio of 0.453, better than its in sample Sharpe of -0.163.

It is still well behind simply holding the same stocks equal weighted, which returned 48.7% over the same period. A strategy that performs noticeably better out of sample than in sample is unusual, and it is a reason for caution rather than celebration, especially given how sensitive that single result turned out to be to the correlation threshold.

6. Efficiency Touchpoint

The slowest step in this pipeline is building the weekly correlation network, since every pair of stocks has to be checked against every other pair. For n stocks, that is n times n minus one over two comparisons, a Big O of O(n^2), and it has to be redone fresh for every week in the sample.

At this universe's size, 56 stocks, that works out to 1,540 pairs a week across 396 weekly networks. The notebook times this directly at 3.6 seconds in total, so at this scale it is not a practical bottleneck.

Once a given week's network exists, computing degree centrality for every stock in it is linear in the number of connections, negligible next to the cost of building the network in the first place.

This stays fast despite growing quadratically because the correlation matrix is computed with a single vectorised pandas call each week, rather than looping over every pair of stocks in plain Python. That keeps the constant factor small even though the number of operations is still of order n squared.

A meaningfully larger universe, in the hundreds or thousands of stocks, would still work with this same approach, but the weekly cost would grow with the square of that size. At that scale, the more efficient path would be to update each week's correlation matrix incrementally from the week before, since consecutive weeks share all but a handful of trading days, rather than recomputing the full 60-day window from scratch each week.

7. Risk Touchpoint

Risk here is measured with the Sharpe ratio and the maximum drawdown, alongside the cumulative return itself. All three are computed separately for the in sample and out of sample periods, rather than blended into one number that would hide how differently the strategy behaved in each.

The strategy's maximum drawdown is worse than either benchmark's in both periods, and so is its annualized return. In sample it lost 57.1% at its worst against an annualized return of -5.9%, compared to a 40.8% drawdown and 28.8% annualized return for the equal weighted basket of stocks, and a 33.9% drawdown and 14.0% annualized return for the S&P 500. Out of sample it lost 18.2% at its worst against a 9.5% annualized return, compared to 14.6% drawdown and 48.9% annualized return for the equal weighted basket, and 9.1% drawdown and 15.7% annualized return for the S&P 500.

The out of sample drawdown is much smaller than the in sample one in absolute terms, which looks like an improvement at first glance. Part of that gap is simply the length of the two windows: the in-sample period runs to roughly 6.5 years against the 1 year held out, so there are far more trading days in which a large drawdown could occur in the first place. Even allowing for that, the out of sample drawdown is still larger than what simply holding the underlying stocks would have produced over the same period. In other words, the strategy does not actually reduce risk relative to its benchmarks in either period, or a meaningful part of its apparent out of sample improvement is really just a much shorter window with less time for things to go wrong.

There is also a concentration risk worth naming. The universe itself is entirely tech and semiconductor stocks, and within that the hubs driving the signal cluster even more narrowly, around semiconductor equipment and analog names like MPWR, KLAC, ENTG, AMAT and LRCX, rather than being spread evenly across the sector. That means a single shock to that specific slice of the chip supply chain, rather than to technology as a whole, can trigger a large share of the strategy's trades at the same time. Nothing in the design spreads that risk across unrelated sectors or hubs, so a downturn concentrated in semiconductor equipment specifically would hit the strategy much harder than it would hit a diversified portfolio.

On the other hand, the strategy carries no leverage. Capital is split equally across whatever trades happen to be open at any time, and since a position was only open on 48% of trading days on average, a meaningful share of the capital sits in cash rather than at risk on any given day. That is a conservative design choice, and it caps the downside at any one point in time. But it also caps how much the strategy could realistically be scaled up: more capital would either have to go into larger positions in the same small set of neighbour stocks, concentrating the risk further, or sit idle, since the number of hub shocks the signal finds each year is fixed by the signal itself rather than by how much money is behind it.

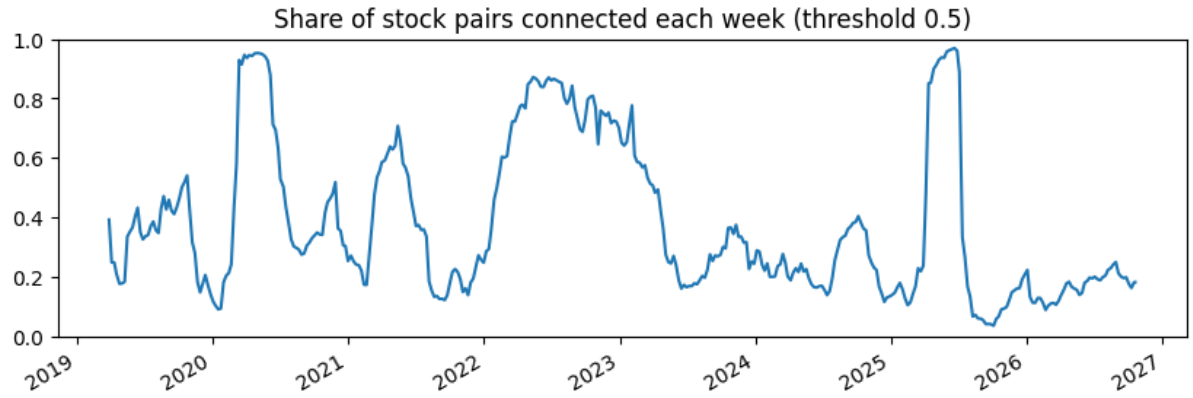


Figure 5 - Share of stock pairs connected each week

Figure 1's right hand panel already showed us that a market wide stress period pushes this network towards saturation. This is shown in the week of 20 June 2025, whose 60=-day window covers the April 2025 tarrif sell-off, when 97% of all possible pairs were linked, as opposed to the 28% median across the full sample (Figure 5). When the network is so dense like this, almost every stock is tied to almost every other stock, so degree centrality stops separating hubs from the rest, and the hub/neighbour distinction of the strategy relies on effectively disappears. A shock to any one name in that kind of week drags in neighbours that are really just the whole market moving together, so every trade the strategy opens becomes the same bet rather than a targeted sector call

A plausible bad scenario brings those two points together. A demand shock specific to semiconductor equipment, such as a major capex pullback by a large chip manufacturer or new export restrictions on equipment sales, would likely move several of the clustered hubs, MPWR, KLAC, ENTG, AMAT and LRCX among them, on the same day, since they share customers and input costs. The strategy would then open several trades at once that are really all the same bet on the same underlying factor, so splitting capital equally across them, as described above, would not actually diversify anything. A loss on one position would likely coincide with losses on the others, concentrating exactly the risk the equal split was meant to spread.

8. Reflection

Overall, the evidence does not support the original hypothesis in this universe. As we have seen from our tests, the lead lag centrality strategy does not beat simply holding the same stocks, in sample or out of sample, once realistic costs are applied. Reporting this honestly, rather than reshaping the hypothesis to fit the result, is the professionally correct approach, and a clean negative result like this still rules out one specific mechanism rather than leaving the question open.

The most valuable finding from this pipeline was not the backtest's return itself, but the lag check in the Sanity Checks section. It showed that connected stocks do move together strongly around a shock, just almost entirely on the same day, rather than with the delayed catch-up the hypothesis relied on.

That result reframes what the correlation network is actually measuring. It is a real, economically sensible map of which stocks share exposure to common sector news. But it is evidence of same day movement, not of the gradual, limited attention driven catch up effect that would have made this a genuinely profitable trading signal.

If this experiment were conducted again, testing for a lag directly, early on, would come before building a full backtest around it. Discovering only at the sanity check stage that the mechanism the strategy depends on barely exists in this sample is a lesson in testing the core assumption first.

A few further extensions would make this work more robust:

- **Survivorship bias**: use a point in time constituents list rather than today's surviving universe, so stocks that were delisted or acquired are included rather than excluded by construction. This was left here as a known limitation rather than fixed within scope.
- **Intraday lag**: this project only ever tested for a catch up across separate trading days. Any delayed reaction could exist within the same day, at the minute or hourly level, which daily data cannot see, so this remains an open question rather than a ruled out one.
- **Broader parameter sensitivity**: only the correlation threshold was stress tested, across 0.4, 0.5 and 0.6, and it was not fully robust there. The window length, hold period, shock size and number of hubs were never tested the same way, so it is unclear whether the result is similarly fragile to those choices too.
- **Realistic slippage modelling**: the 10-basis point cost is applied once per trade rather than split across entry and exit and does not scale with position size or the liquidity of the specific neighbour stock being traded. A size and liquidity dependent slippage model would be a more realistic cost assumption.

References

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